Optimal objective and error

ML ingredients

When solving a task with ML **we need to**:

- Define the **task**
- **Collect** (annotated) **data**
- **Define** a **family of models** to solve the task → **Hypothesis space**
- **Find** the **model** that **minimizes** the **error** → **Learning algorithm**
- **Define** an **error function to measure** how well a model fits the data → **Optimal target**

Task

A **task** is a **set of functions** that **can** potentially **solve a problem**

Each function assigns to each input $x \in X$ an output $y \in Y$

$$f : X \to Y$$

Data

Data is the **information about the problem** to be solved

It has the **form of a** (probability) **distribution**

- Usually referred to as p_{data}
- Is **typically unknown**, but we can sample it
- **Training**, **validation** and **test** set **share it**

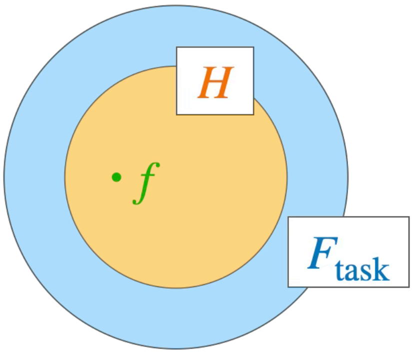
⚠️ **The failure** of ML algorithms is **often caused** by a **bad selection of training** examples

Model and Hypothesis Space

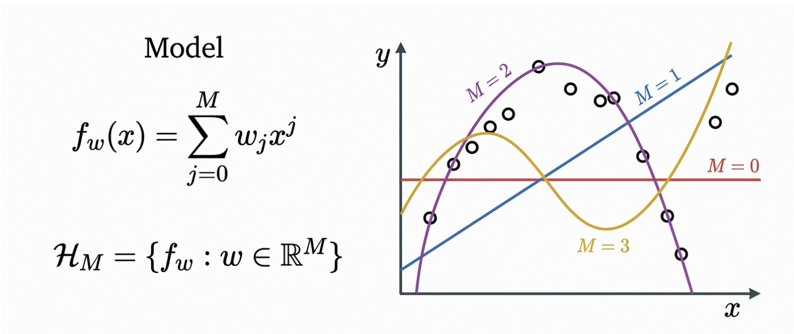
A model is the implementation of a function $f \in F_{\text{task}}$, where F_{task} is a set of candidate functions

A set of models forms an hypothesis space $H \in F_{\text{task}}$

→ **The learning algorithm seeks a solution within the hypothesis space**



Set representation



Example: polynomial curve fitting

Objective

The search of the best model is based on **minimizing the error function**

We need to consider 3 cases:

Ideal target

$$f^\star \in \arg\min_{f \in F_{\text{task}}} E(f; p_{\text{data}})$$

Where:

- $f^\star$ is a function or a set of functions
- $E(f; p_{\text{data}})$ is the generalization error
- $\arg\min_{f \in F_{\text{task}}}$ indicates the space of functions in F_{task} with minimum generalization error

In particular:

- It **measures the error on the whole available data**

Has the following problems:

- **The search space is too large** (and may contain **functions** that **can't be implemented**)
- Is **NOT computable** as p_{data} **is unknown**

Feasible target

$$f_H^\star \in \arg\min_{f \in H} E(f; p_{\text{data}})$$

Where:

- $f_H^\star$ is a function or a set of functions
- $E(f; p_{\text{data}})$ is the generalization error
- $\arg\min_{f \in H}$ indicates the space of functions in H with minimum generalization error

In particular:

- It **measures the error on the whole available data**
- The **search space is restricted**, compared to the ideal target

Has the following problems:

- Is **NOT computable as p_{data} is unknown**

Actual target

$$f_H^\star(T) \in \arg\min_{f \in H} E(f; T)$$

Where:

- $f_H^\star(T)$ is a function or a set of functions
- $E(f; T)$ is the training error
- $\arg\min_{f \in H}$ indicates the space of functions in H with minimum training error

In particular:

- It **measures the error only on the training set T**
- The **search space is restricted**
- Is **computable, as T is known**

⚠️ The learning algorithm solves **the optimization problem targeting** $f_H^\star(T)$, but **might end up in a different result** $\hat{f}_H^\star(T)$ (because of local mimima, saddle points, …)

Error functions

Used to measure **how likely a model fits the data**

Can be written in terms of a **pointwise loss** $L(f; (x_i, y_i))$

Generalization error

$$E(f; p_{\text{data}}) = \mathbb{E}_{(x,y)\sim p_{\text{data}}}[L(f;(x,y))]$$

- p_{data} is **unknown**, as a cons **the generalization error too**
- $\mathbb{E}_{(x,y)\sim p_{\text{data}}}$ is the **expectation calculated from values sampled from** p_{data}

Training error

$$E(f;T) = \frac{1}{N}\sum_{i=1}^{N} L\big(f;(xi, yi)\big)$$

- Is the **average error value within the** T **set**

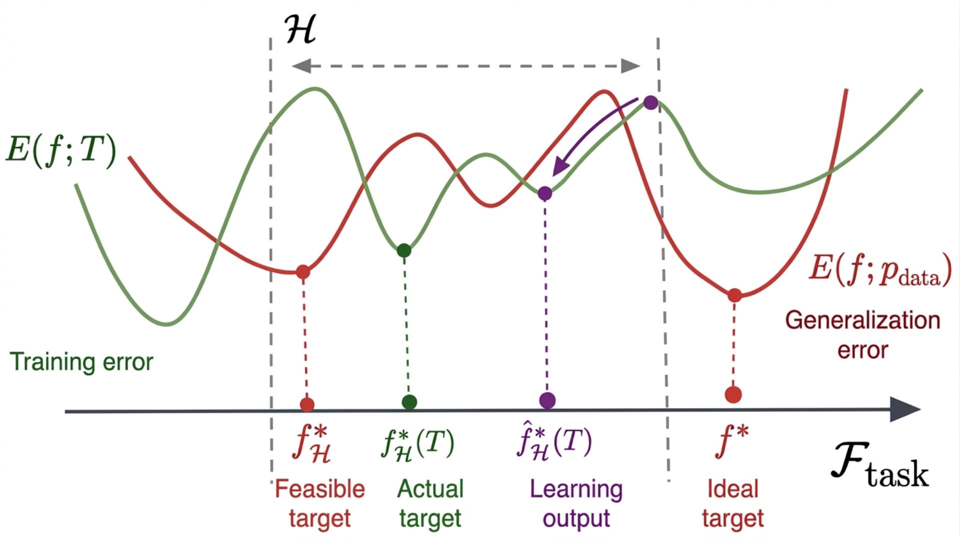
💡 The **aim** is to **minimize both the training error** and the **generalization error**

Approximation error

- Is the **error induced by the use of the hypothesis space** H instead of the whole F_{task}
  - Is the generalization error, taking into account H only

Irreducible error

- Is an error that **can't be reduced by choosing a different model**
- Occurs because of:
  - Natural variability and randomness of the system
  - Deterministic factors that can't be observed
  - → Represents the **fundamental uncertainty in the problem**



Underfitting and Overfitting

They indicate **how much a model fits** the training data

Underfitting

The **model is too simple** to capture the underlaying patterns:

- **High training error**
- **High generalization error**

→ **Poor performance**, on both training and test data

Overfitting

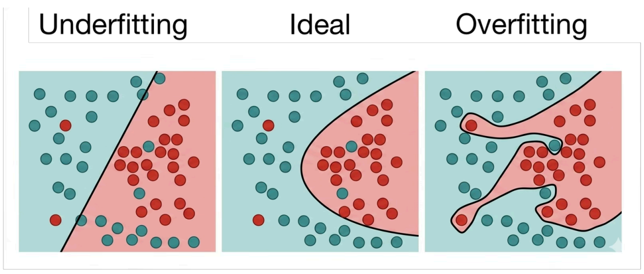
The **model is too complex** and sticks to training data too much:

- **Low training error**
- **High generalization error**

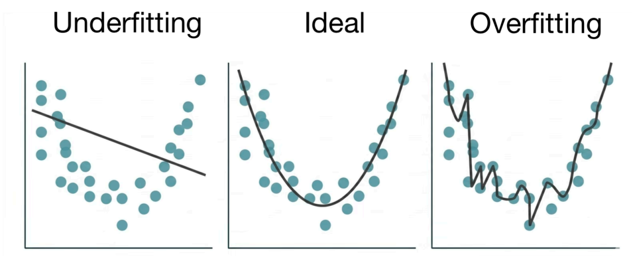
→ **Performs** well on training data, but **poorly on testing** and **unseen data**

💡 In the **ideal** situation:

- **Low training error**
- **Low generalization error**
- **Small gap between training and generalization error** → high performance in both cases



Classification



Regression

Capacity

Indicates the **ability of a model to fit a wide variety of functions**

Is controlled by:

- Number of parameters set
- Model architecture
- Features dimensionality

Low capacity

The **model is limited** to **simple patterns**

- Is associated to **underfitting**

High capacity

The **model** can represent **very complex patterns**

- Is associated to a **higher risk of overfitting**

💡 In the **ideal** situation **a mid-level capacity occurs**, in order to gain an ideal fitting

Improving generalization

To improve the generalization of the model, a few approaches exist

Regularization

Consists in **modifying the training error function of a** $\Omega(f)$ **factor**
(= a factor that grows at least as much as f)

- **The more complex** a model is, **the higher the overfitting** is
- **Penalizes complex solutions**

$$E_{\text{reg}}(f;T) = E(f;T) + \lambda\Omega(f)$$

Augmenting the training set

As the **number of samples increases**,
the **similarity between** the **training** and the **generalization error increases**

$$E(f;T) \rightarrow E(f;p_{\text{data}}) \text{ as } n \rightarrow \infty$$

Combining predictions from multiple models

Combining predictions from multiple models that are decorrelated improves generalization